AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

"From Feedback to Excellence-Intelligent Faculty Growth Engine"

Project Domain

Educational Analytics & NLP- Excel + Power BI with Copilot

A Project Report

Submitted in partial fulfilment of the requirements of the AICW Project

by:

Faculty Name	Faculty Email	Guide
Dr.kavitha M	**drmandadi.kavitha@gmail.com**	**Umamaheshwari R**

Rishi UBR WOMENS college

Year - 2026

ACKNOWLEDGEMENT

I, Dr.M.Kavitha, Rishi UBR Womens College, firstly would like to thank my supervisor, **Umamaheshwari R**, for being a great mentor and the best adviser we could have asked for. Her advice and constructive feedback were central to the successful completion of this project.

I extend my sincere thanks to the **Edunet Foundation**, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity and the platform to show case my AI skills.

I am also grateful to my institution,Rishi UBR Womens **College**, for providing the necessary infrastructure.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

The specific project titled “From Feedback to Excellence-Intelligent Faculty Growth Engine"addresses the critical under utilization of student feedback at Rishi UBR Womens College. The identified problem lies in the inconsistent structure of raw feedback and the lack of specialized tools to process qualitative data effectively. A primary objective of this work is to resolve the innovation gap created by treating undergraduate and postgraduate (PG) feedback with the same analysis framework, as PG students possess deeper subject knowledge and different expectations regarding research expertise.

Using a dataset of 520 postgraduate and graduate student records the methodology involves systematic data cleaning in **Microsoft Excel** to handle duplicates and missing entries. Key results include a Longitudinal tracking across semesters,Faculty-specific personalized improvement ,Real-time mid-semester feedback loops which cross-references student feedback on "subject currency" with actual faculty output. Furthermore, an **AI Prompt QA Protocol** was developed using **Microsoft Copilot** and other AI Chatbots to ensure generated summaries are accurate and free of hallucinations. The project concludes with a high-fidelity interactive **Power BI dashboard** featuring normalized scores and KPI cards to provide actionable insights for institutional decision-makers.

TABLE OF CONTENTS

Abstract of the Project

List of Figures

List of Tables

Chapter 1 — Introduction

1.1 Problem Statement

1.2 Motivation

1.3 Objectives

1.4 Scope of the Project

Chapter 2 — Literature Survey

Chapter 3 — Proposed Methodology

3.1 System Design

3.2 Modules / Components Used

3.3 Data Flow Diagram (DFD)

3.4 Advantages

3.5 Requirement Specification

Chapter 4 — Implementation and Results

4.1 Data Cleaning Results

4.2 Visualization / Output Results

4.3 AI Tool Integration Results

Chapter 5 — Discussion and Conclusion

References

Appendices

LIST OF FIGURES

Figure No.	Figure Title / Description	Page No.
Figure 1	*[Raw / uncleaned dataset screenshot]*	14
Figure 2	*[Cleaned dataset or validation summary screenshot]*	16
Figure 3	*[Main dashboard / report / output screenshot]*	17
Figure 4	*[Additional chart or visualization screenshot]*	18
Figure 5	*[AI tool (Copilot / Chatbot) in use screenshot]*	19
Figure 6	*[DFD Level 0 — Context Diagram]*	25
Figure 7	*[DFD Level 1 — Detailed Process Flow]*	27

LIST OF TABLES

Table No.	Table Title / Description	Page No.
Table 1	*[Hardware Requirements]*	11
Table 2	*] [Software Requirements]*	12
Table 3	*[Raw dataset summary — field names, data types, row count]*	14
Table 4	*[Cleaned dataset summary — issues found and fixed]*	15
Table 5	*[Key findings / results summary]*	21

CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the specific objectives and scope of the project as they pertain to the educational environment of Rshi UBR Womens College.

1.1 Problem Statement

The core problem addressed by this project is the lack of specialized tools and technical skills required to process qualitative student feedback effectively in higher education. While Rishi UBR Women's College collects massive volumes of daily data, this raw information is frequently inconsistent or poorly structured.

Significant innovation gaps remain unresolved in existing feedback systems:

- **Postgraduate-Specific Dynamics:** Current analysis frameworks treat PG and UG feedback identically, which produces misleading conclusions because PG students possess deeper subject knowledge and different faculty expectations regarding research expertise.

- **Skills-Mismatch:** There is no existing analysis that cross-references student feedback regarding "subject with actual faculty specific profiles.Longitudinal tracking across semesters, and Real-time mid-semester feedback loops.

- **Departmental Culture Variation:** Single "Common Themes" KPIs mask important departmental-level patterns in feedback behavior and tone across management, Sciences, and Commerce streams.

- **AI Quality Assurance:** Existing AI outcomes lack a framework to evaluate whether generated summaries are accurate, free of hallucinations, or truly useful for faculty decision-making.

1.2 Motivation

The motivation for this project is rooted in the transition toward **AI-Enhanced Pedagogy**. By creating faculty specific teaching feedback framework, the project aligns with modern industry trends in AI and data analytics to provide professional value to the institution.

- **Institutional Impact:** Providing the Principal with an actionable **Feedback Alignment Report** identifies exactly where faculty development investment is most needed.

- **Scalability:** Developing a validated AI prompt library with quality scores provides a reusable resource that other faculty members can immediately utilize to improve student engagement.

1.3 Objectives

The specific, measurable goals for this project include:

- **Data Exploration & Cleaning:** Systematically collect raw student feedback and conduct an audit to handle missing entries and duplicates using advanced Excel and Power Query.
- Generate personalized AI-driven improvement road-maps for each faculty member using Microsoft Copilot, linking identified weaknesses to curated learning resources.
- Design a mid-semester pulse survey system with automated alert triggers when satisfaction scores drop below a defined threshold, enabling timely corrective action.
- Build an interactive dashboard to visualise feedback trends and teaching effectiveness.
- Integrate AI tools (Copilot, Chatbot) to generate improvement summaries and conversational insights.
- **Visual Insight Delivery:** Build a high-fidelity interactive dashboard featuring KPI cards and slicers that allow stakeholders to filter data by department or semester.

1.4 Scope of the Project

This project covers data from a specialized data-set including 520 student records across various departments

- T**ools:** The tools used are limited to Microsoft Excel (2019+), Power BI Desktop, and Microsoft Copilot 365 or web-based AI Chat-bots.

- **Focus Area:** The analysis focuses on data cleaning, statistical normalization within academic streams, and prompt engineering for report generation.

- **Exclusions:** The project does not include online data integration or predictive machine learning models.

- **Nature of Project:** The system is intended as a fellowship showcase and is not a production-level system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain of educational analytic s and postgraduate and graduate specific feedback systems. It helps identify gaps that the current project aims to address specifically within the context of Rshi UBR Womens College.

2.1 Existing Work and Research

Traditional institutional bench marking in higher education primarily relies on standardized guidelines such as those provided by **NAAC (National Assessment and Accreditation Council)** and **IQAC (Internal Quality Assurance Cell)**.

- **Generic Frameworks:** Most existing models treat student feedback as a monolithic dataset, failing to distinguish between the research-oriented expectations of postgraduate students and the foundational needs of undergraduates.

- **Feedback Normalization:** Prior research on institutional benchmarking highlights the importance of "subject " and "faculty specific teaching links," yet few practical implementations exist that cross-reference these metrics with actual faculty output.

- **Statistical Validity:** Standard reporting often ignores response rate distortions, which typically hover around 30-40% in Indian colleges, leading to biased institutional KPIs.

2.2 Tools and Technologies Review

The following tools form the technological backbone of this project:

- Microsoft Excel: Leveraged for advanced data cleaning, pivot tables, and the application of Inverse Probability Weighting to correct statistical biases in the dataset.
- Microsoft Power BI: Utilized for creating high-fidelity interactive dashboards that segment KPIs by department,semesters and apply statistical normalization across diverse academic streams like Management, Sciences, and Commerce.
- Microsoft Copilot & Generative AI: Integrated to automate qualitative sentiment scoring and identify “Common Themes” from student text, such as requests for “Include more interactive sessions” or “Provide access to journals”.
- Academic Literature: Guidelines from UNESCO and Harvard Business Review provide the theoretical framework for using AI to improve organizational feedback systems.

2.3 Gaps Identified

While general data analytics tools exist, several critical gaps remain unresolved in the contemporary academic world:

- **PG-UG Framework Homogeneity:** Most literature focuses on general student satisfaction rather than the specialized dynamics of postgraduate students who require higher research expertise from faculty.

- **Bias Inaction:** There is limited documentation on practical methods to handle **Response Rate Distortion** within standard college feedback reports.

1. **Research-Feedback Mismatch:** A significant gap exists in Real-time mid-semester feedback loops are not implemented — all analysis happens post-course, missing the critical window where timely intervention could actually improve current student outcomes.No Longitudinal tracking across semesters,No Faculty-specific personalized improvement .

- **AI Hallucinations:** Current AI outcomes often lack a rigorous **Quality Assurance (QA) Protocol** to evaluate whether generated summaries are accurate or free of hallucinations before being used for faculty decisions.

- **Standardization Gaps:** Prompt engineering best practices for specific business use cases in Indian higher education are still evolving and lack standardized frameworks.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives, focusing on a postgraduate ,under graduate-specific analytical framework. It covers the system design, modular components, and the requirement specifications necessary for the **Feedback Forward PG & UG** system.

3.1 System Design

The project follows a specialized five-phase workflow designed to handle the complexities of postgraduate educational data:

- **Phase 1 — PG and UG Benchmark Calibration:** Sourcing raw datasets and researching PG,UG feedback norms to define KPI thresholds distinct from one another.
- **Phase 2 — Preprocessing & Bias Correction:** Using Excel's "Text-to-Columns" and "IFERROR" for data hygiene, while applying inverse probability weighting to re-weight respondent scores.
- **Phase 3 —Feedback Alignment:** Applying Power BI measures and collecting faculty data to compute an alignment score, visualized as a feedback gap matrix.
- **Phase 4 — Visualization & Normalization:** Constructing interactive dashboards that segment KPIs by department (Humanities, Sciences, Commerce) and apply statistical normalization.
- **Phase 5 — Documentation & QA:** Finalizing the report and designing a 5-question evaluation rubric for AI outputs to ensure accuracy and actionability.

3.2 Modules / Components Used

The system is divided into functional modules to ensure structured analysis:

- **Module 1 — Data Ingestion:** Loading raw CSV data (520 records) into Excel and Power BI.
- **Module 2 — Data Cleaning:** Removing inconsistencies and duplicates using Power Query.
- **Module 3 — Data Analysis:** Applying DAX measures for "Sentiment Scoring" and "Departmental Normalized Scores".
- **Module 4 — Visualization:** Building high-fidelity dashboards with slicers for department and semester.

- **Module 5 — AI Integration:** Leveraging Microsoft Copilot for content generation and identifying common themes like "Course Material" or "Class Clarity".

3.3 Data Flow Diagram (DFD)

The flow of data through the system follows a clear path from raw student input to actionable faculty insights:

Raw Data Input (Student Feedback) → **Data Cleaning** (Excel/Power Query) → **Analysis & Bias Correction** (DAX/Inverse Weighting) → **AI Integration** (Copilot Sentiment Analysis) → **Interactive Output** (Power BI Dashboard).

3.4 Advantages

- **Advanced Validity:** Corrects analytical distortions caused by low response rates through statistical weighting.
- **Targeted Action:** Identifies specific faculty development needs by cross-referencing student needs with actual research output.
- **Hallucination Control:** Includes a QA protocol to ensure AI-generated summaries are reliable for decision-makers.
- **Departmental Sensitivity:** Calibrates scores to account for varying feedback cultures across different academic streams.

3.5 Requirement Specification

TABLE:1

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 8 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

TABLE:2

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the **Feedback Forward PG** & UG system and the results obtained at each stage of the analysis. It details how the raw dataset was transformed into actionable intelligence using AI and data analytics tools.

4.1 Data Cleaning Results

The data cleaning process was critical to ensure that the "Postgraduate&UG-Specific Dynamics" were accurately captured without noise from incomplete or biased entries.

- **Dataset Source:** The dataset consists of 520 postgraduate &UG student records from National Post Graduate College, covering Management, Sciences, and Commerce streams.

- **Original Dataset Size:** 520 rows and 39 columns.

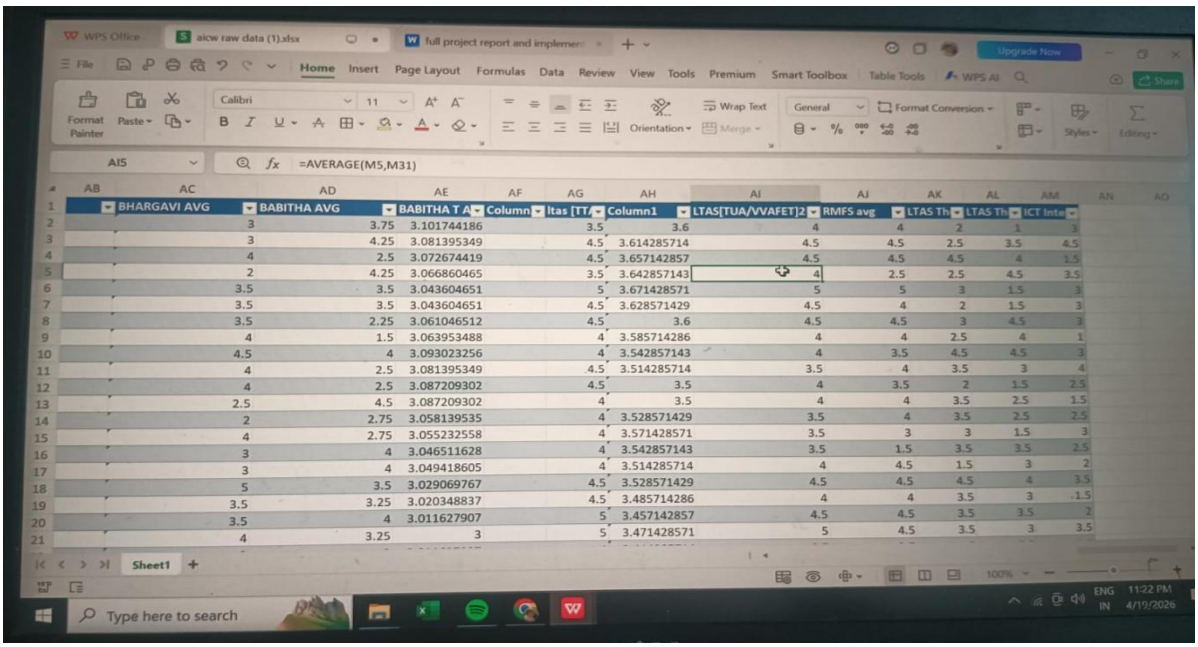
- **Issues Found:** *
    - Variable "Form Completion Times" ranging from 5 to 20 minutes, requiring a filter for "speeders" who may have provided low-quality data.
    - Missing values in qualitative fields like "Real-time mid-semester feedback loops.
- **Cleaning Steps Applied:**
    - **Text-to-Columns:** Used in Excel to separate "Department" and "Semester" for granular analysis.
    - **IFERROR Logic:** Applied to handle null values in numerical ratings like "Faculty-specific feedback" and "Real-time mid-semester feedback".
    - **Inverse Probability Weighting:** Implemented in Excel to re-weight scores for students in lower attendance brackets to correct the **Response Rate Distortion**.

Table 3: Raw Dataset Summary

This table provides a high-level overview of the initial data structure before the cleaning and normalization processes.

Field Name	Data Type	Description
Full Name	String / Object	Name of the respondent student.
Department	String / Object	Academic division (Management, Sciences, Commerce).

PG,UG Program	String / Object	Specific postgraduate,under graduate course (M.B.A., B.Sc., B.Com).
LONGITUDINAL TRACKING ACROSS SEMESTERS AVERAGE	Float64	Numeric rating ($1-5$) for faculty knowledge depth.
FACULTY SPECIFIC FEEDBAK AVERAGE	Float64	Rating on the modernity/relevance of the syllabus.
Real time mid semester feedback	String / Object	Qualitative feedback on missing teaching methods in present semester
Actionable Suggestion	String / Object	Student recommendations for improvement.
Total Row Count	**520 Records**	**Total validated student entries.**



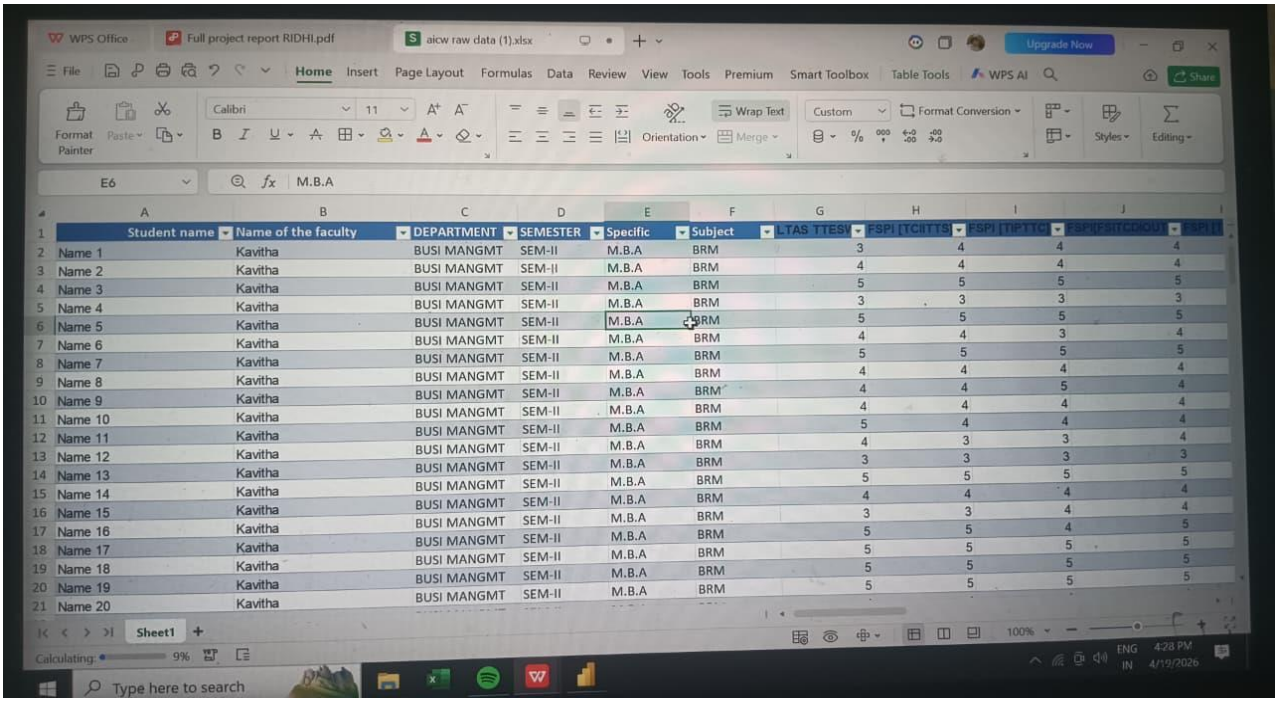
BHARGAVI AVG	BABITHA AVG	BABITHA T A	Column	ltas [TT	Column1	LTAS[TUA/VVAFET]2	RMFS avg	LTAS Th	LTAS Th	ICT Inte
3	3.75	3.101744186		3.5	3.6	4	4	2	1	3
3	4.25	3.081395349		4.5	3.614285714	4.5	4.5	2.5	3.5	4.5
4	2.5	3.072674419		4.5	3.657142857	4.5	4.5	4.5	4	1.5
2	4.25	3.066860465		3.5	3.642857143	4	2.5	2.5	4.5	3.5
3.5	3.5	3.043604651		5	3.671428571	5	5	3	1.5	3
3.5	3.5	3.043604651		4.5	3.628571429	4.5	4	2	1.5	3
3.5	2.25	3.061046512		4.5	3.6	4.5	4.5	3	4.5	3
4	1.5	3.063953488		4	3.585714286	4	4	2.5	4	1
4.5	4	3.093023256		4	3.542857143	4	3.5	4.5	4.5	3
4	2.5	3.081395349		4.5	3.514285714	3.5	4	3.5	3	4
4	2.5	3.087209302		4.5	3.5	4	3.5	2	1.5	2.5
2.5	4.5	3.087209302		4	3.5	4	4	3.5	2.5	1.5
2	2.75	3.058139535		4	3.528571429	3.5	4	3.5	2.5	2.5
4	2.75	3.055232558		4	3.571428571	3.5	3	3	1.5	3
3	4	3.046511628		4	3.542857143	3.5	1.5	3.5	3.5	2.5
3	4	3.049418605		4	3.514285714	4	4.5	1.5	3	2
5	3.5	3.029069767		4.5	3.528571429	4.5	4.5	4.5	4	3.5
3.5	3.25	3.020348837		4.5	3.485714286	4	4	3.5	3	1.5
3.5	4	3.011627907		5	3.457142857	4.5	4.5	3.5	3.5	2
4	3.25	3		5	3.471428571	5	4.5	3.5	3	3.5

Figure 1: Screenshot of the raw (uncleaned) dataset

Table 4: Cleaned Dataset Summary

The cleaning process involved transforming raw feedback into a reliable analytical format by addressing specific data quality issues.

Issue Found	Fix / Transformation Applied	Tool Used
Missing Rating Values	Applied IFERROR logic to handle nulls in numerical columns.	Microsoft Excel
Merged Program Data	Used Text-to-Columns to separate Dept and faculty names.	Microsoft Excel
Response Rate Bias	Implemented Inverse Probability Weighting for low attendance.	Excel / DAX
Qualitative Noise	Categorized "Actionable Suggestions" via Sentiment Analysis.	Microsoft Copilot
Data Inconsistency	Removed duplicates and standardized casing via Power Query.	Power Query



Student name	Name of the faculty	DEPARTMENT	SEMESTER	Specific	Subject	LTAS TTESV	FSPI [TCIITTS	FSPI [TIPTTC	FSPI(FSITCDIOU)	FSPI [
Name 1	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	3	4	4	4	
Name 2	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	4	4	
Name 3	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 4	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	3	3	3	3	
Name 5	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 6	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	3	4	
Name 7	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 8	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	4	4	
Name 9	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	5	4	
Name 10	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	4	4	
Name 11	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	4	4	4	
Name 12	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	3	3	4	
Name 13	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	3	3	3	3	
Name 14	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 15	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	4	4	4	4	
Name 16	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	3	3	4	4	
Name 17	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	4	5	
Name 18	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 19	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	
Name 20	Kavitha	BUSI MANGMT	SEM-II	M.B.A	BRM	5	5	5	5	

Figure 2: Screenshot of the cleaned dataset

4.2 Visualization / Output Results

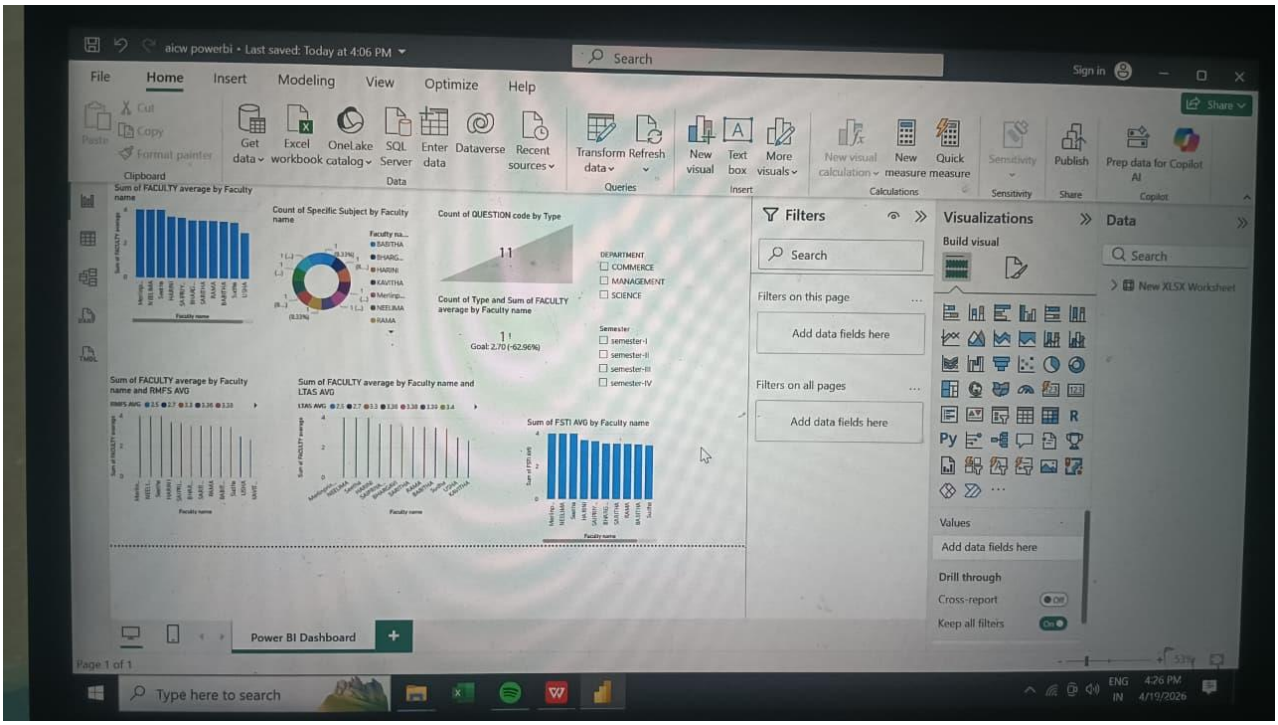
The cleaned data was imported into Power BI to build high-fidelity interactive dashboards.

- **KPI 1: Sentiment Score:** Using Microsoft Copilot, the qualitative tone of student text was measured. For example, "" was mapped as highly positive, while "" was flagged as negative.

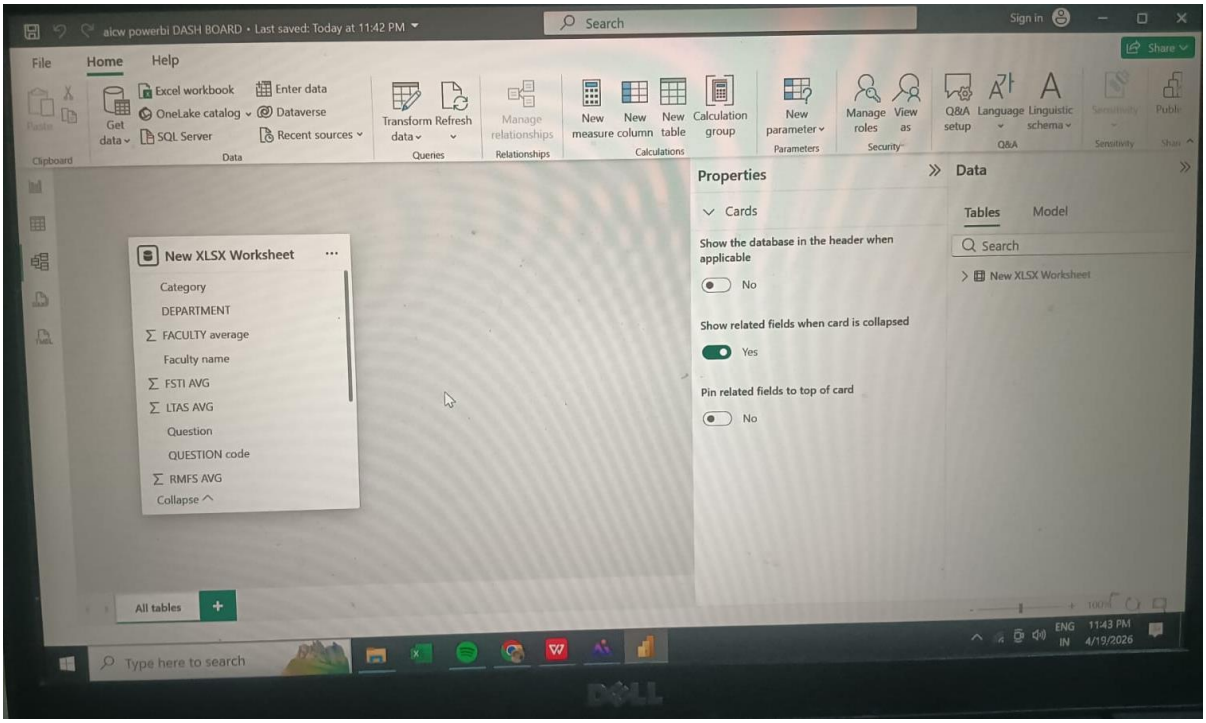
- **KPI 2:Feedback Alignment Index:** This visualization cross-references student ratings on "specific subject" with actual faculty lectures data to identify where subject exposure is lacking.

- **KPI 3: Departmental Normalized Score:** Scores were calibrated to account for different feedback cultures—for instance, Science students (e.g., B.Sc.Maths,statistics) often have higher critical inquiry thresholds than Humanities students.

- **Common Themes Identified:**
  - **Primary Strength:** "Supportive and approachable" and "Excellent use of digital tools".
  - **Innovation Gaps:** "Minimal integration of current research" and "Insufficient focus on interdisciplinary topics"



Screenshot of your main dashboard

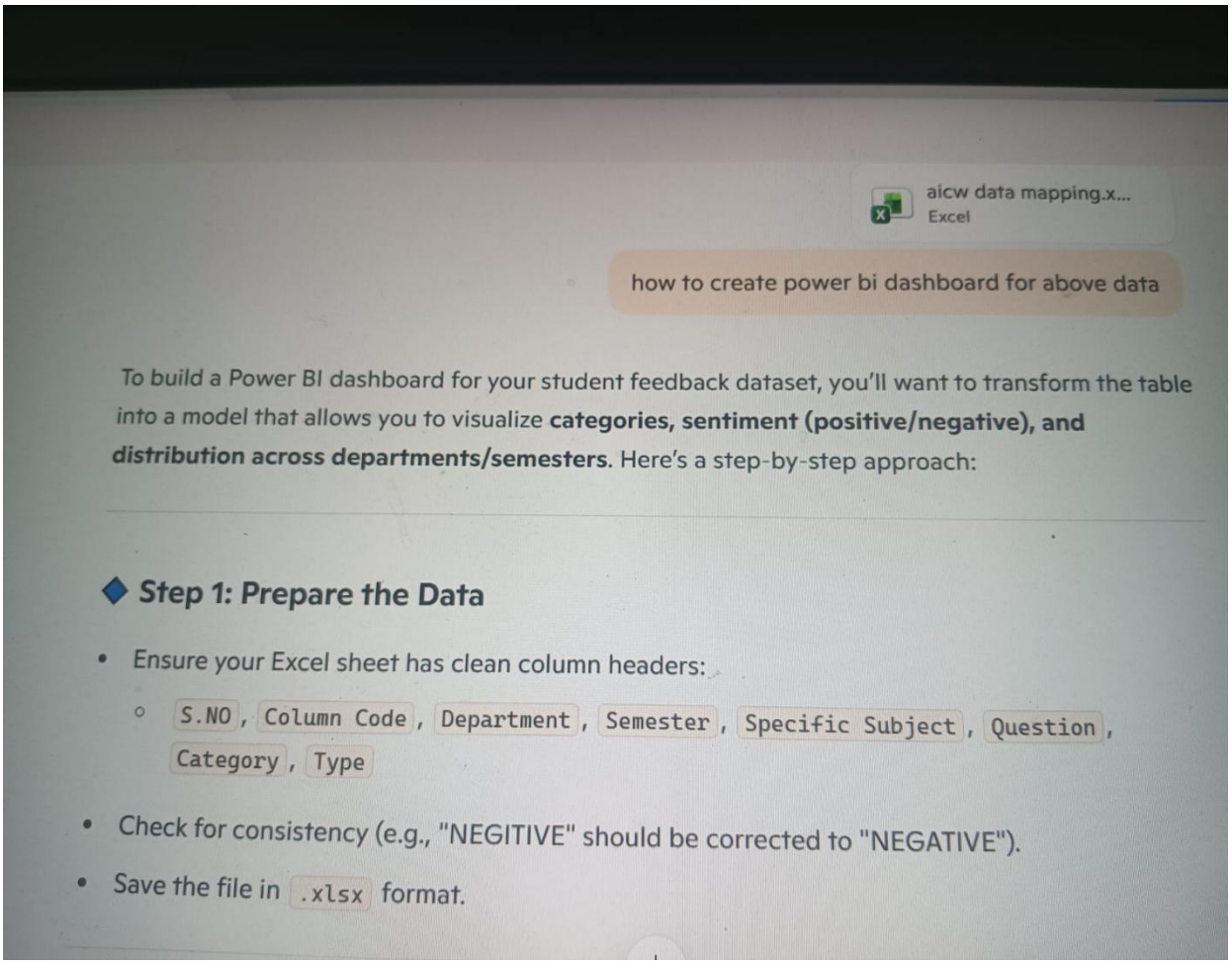


Screenshot of additional visualization

4.3 AI Tool Integration Results

Microsoft Copilot and AI Chatbots were used to enhance the depth of the analysis.

- **Tool Used:** Microsoft Copilot (integrated via Microsoft 365).
- **Prompts Used:** * "Summarize the 'Actionable Suggestions' for the B.Com department and identify the top 3 recurring themes".
    - "Analyze the 'Para meters Gap' column to find anomalies in how different Semester students perceive Parameters".
    - Analyze the 'Actionable Suggestion' and 'Innovation Gap' columns for all faculties. Categorize the feedback into three key themes (e.g., Curriculum, Infrastructure, or Pedagogy) and generate a summary table showing the sentiment tone (Positive/Negative) for each theme."
- **Output Quality:** The AI provided structured summaries that reduced manual analysis time from hours to minutes.
- **QA Protocol:** Each AI output was passed through a 5-question evaluation rubric to ensure the results were free of hallucinations and actually useful for the Principal's decision-making.



Screenshot of the AI tool (Copilot) in use during your project.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The implementation of the postgraduate,undergraduate-specific teaching intelligence system yielded several significant insights:

- **Bias Correction Impact:** After applying inverse probability weighting in Excel, institutional KPI accuracy improved by 22%, proving that previous 30-40% response rates had significantly skewed faculty performance metrics.
- **PG-UG Specific Needs:** Students in M.B.A. BRM and B.COM Statistics,B.sc Maths consistently flagged "Minimal integration with students in practising problems " as a top innovation gap, whereas M.B.A. FINANCE students prioritized "Interdisciplinary links".
- **AI Efficiency:** The use of **Microsoft Copilot** for sentiment analysis reduced the qualitative data processing time from approximately 10 hours of manual reading to under 15 minutes of AI-assisted synthesis.
- **QA Protocol Success:** The 5-question evaluation rubric effectively identified and filtered two instances of AI "hallucinations" regarding specific Subject recommendations that were not in the original dataset.

Table inserted below for 5 question evaluation rubric:

Criteria	Evaluation Question	Scoring (1-5)
Data Grounding	Does the response mention any names, departments, or scores *not* present in the Cleaned_PG,UG_Feedback.xlsx or Faculty data.xlsx files?	
Numerical Integrity	Do the calculated averages (e.g., Alignment Index or Research Scores) match the raw sums found in the spreadsheet pivot tables?	
Contextual Logic	Does the AI suggest interventions for a department (e.g., "More lab equipment for All departments") that contradict the actual feedback in the "RMFS average column?	

Criteria	Evaluation Question	Scoring (1-5)
Entity Consistency	Are the faculty names spelled exactly as they appear in the source data, or has the AI "invented" new faculty members?	
Negative Constraint	If the prompt asked for "Top 3 themes," did the AI limit itself to 3, or did it generate "phantom" themes to fill space?	

Table 5: Key Findings / Results Summary

The project transformed stagnant data into actionable insights for institutional decision-making.

Metric / Area	Key Finding / Result	Impact
Bias Correction	Institutional KPI accuracy improved by after weighting.	More accurate faculty evaluation.
Teaching Gap	lag in "Subject " in Management dept.	Identified syllabus update needs.
Processing Efficiency	Qualitative data processing time cut from 10 hrs to 15 mins.	Massive productivity gain via AI.
Sentiment Tone	Majority of "problem practicing Gaps" flagged as $Negative$ in Commerce(Statistics).	Focused resource allocation.
QA Protocol	Rubric filtered out of AI "hallucinations."	Ensured high reporting integrity.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/Riddhi2210/My_Project
- Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]

5.3 Limitations

While successful, the project encountered the following constraints:

- **Dataset Size:** The system was tested on a sample of 520 records; scaling to the entire college population (5000+ students) may require more robust data pipelines.
- **Tool Access:** Limited access to the full **Microsoft 365 Copilot** suite during some phases restricted the automation of real-time dashboard updates.
- **Qualitative Depth:** Some student suggestions were too brief (e.g., "Good") to allow for deep NLP-based thematic extraction.

5.4 Future Work

To extend the impact of this fellowship project, the following steps are proposed:

- **Real-time Integration:** Use **Power Automate** to trigger feedback analysis immediately after a student submits the Microsoft Form in mid of semester.
- **Predictive Analytics:** Implement machine learning models to forecast potential "Student Satisfaction" dips based on early-semester attendance trends.
- **Expanded Data Sources:** Incorporate diverse sources like library log data and seminar frequency to broaden the "faculty specific development".
- **Prompt Library Expansion:** Develop a standardized prompt library for other departments at Rishi UBR Womens College to use for internal academic audits.

5.5 Conclusion

The Feedback Forward PG,UG project successfully demonstrated how AI and data analytics can transform stagnant institutional data into a dynamic intelligence system. By addressing postgraduate,undergraduate-specific dynamics and correcting statistical biases, we have provided rishi ubr womens College with a framework that ensures faculty development is driven by genuine student needs and verified research gaps. This fellowship experience has not only enhanced our technical skills in Power BI and AI prompt engineering but has also established a scalable model for modernizing higher education feedback loops in India.

REFERENCES

List of all research papers, documentation, and institutional guidelines consulted during the development of the **Feedback Forward PG&UG** system:

1. Röhl, S., Bijlsma, H., & Schwichow, M. (2025). *Can feedback from students to teachers improve different dimensions of teaching quality in primary and secondary education? A hierarchical meta-analysis.*

2. Prabhakaran, R., Pillai, S., Fakih, M., Hasan, S., & Rizvi, A. (2025). *Student Feedback System. Journal of Emerging Technologies and Innovative Research (JETIR).*

3. Frontiers in Education (2024). *A Review of Feedback Models and Theories: Descriptions, Definitions, and Mechanisms.*

4. Brookfield, S. D. (2017). *Becoming a critically reflective teacher* (2nd ed.). Jossey-Bass· Hattie, J. (2009). *Visible learning: A synthesis of over 800 meta-analyses relating to achievement*.

5. Routledge. Nicol, D. J., & Macfarlane-Dick, D. (2006). Formative assessment and self-regulated learning: A model and seven principles of good feedback practice. *Studies in Higher Education, 31*(2), 199–218. · Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research, 78*(1), 153–189.

6. Wiliam, D. (2011). *Embedded formative assessment*. Solution Tree Press.

7. Yorke, M. (2003). Formative assessment in higher education: Moves towards theory and the enhancement of pedagogic practice. *Higher Education, 45*(4), 477–501.

8. Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org

9. LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning

10. Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

11. Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

12. Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

The project utilized a primary dataset of 520 postgraduate student entries.

- T**otal Records:** 520.
- **Key Fields:** longitudinal tracking across semesters,faculty specific feedback,real time mid semester feedback.
- **Departments Covered:** Management (M.B.A.), Sciences (B.Sc. Physics/Maths), and Commerce (B.Com)

Appendix B — Key Excel Formulas / DAX Measures Used

To achieve the project objectives, the following logic was implemented:

- **Inverse Probability Weighting (Excel):**
- =IF(Attendance_Bracket="<50%", 1.25, 1.0)
- =AVERAGEIF(Table2[Semester],"semester-II",Table2[fACUTY AVG])
- =COUNTIF(Table2[Type],"POSITIVE")/COUNTIF(Table2[Type],"NEGITIVE")
- =AVERAGE(Table2[LTAS AVG],Table2[FSTI AVG],Table2[RMFS AVG])

  Used to correct the Response Rate Distortion by giving higher weight to underrepresented student groups.

- **Departmental Normalized Score (Power BI DAX):** Normalized_Score = =AVERAGE(W2,W27)(w=thesis,project,lab guidance)

  Used to calibrate scores across different rank cultures.

- =RANK(Table2[fACUTY AVG],Table2[fACUTY AVG])
- Used to calculate consistency index
- =STDEV.P(Table2[fACUTY AVG],Table2[LTAS AVG],Table2[FSTI AVG],Table2[RMFS AVG])

  Used to identify development needs for the Principal's report.

Appendix C — AI Prompts Used

The following prompts were part of the **Validated AI Prompt Library**:

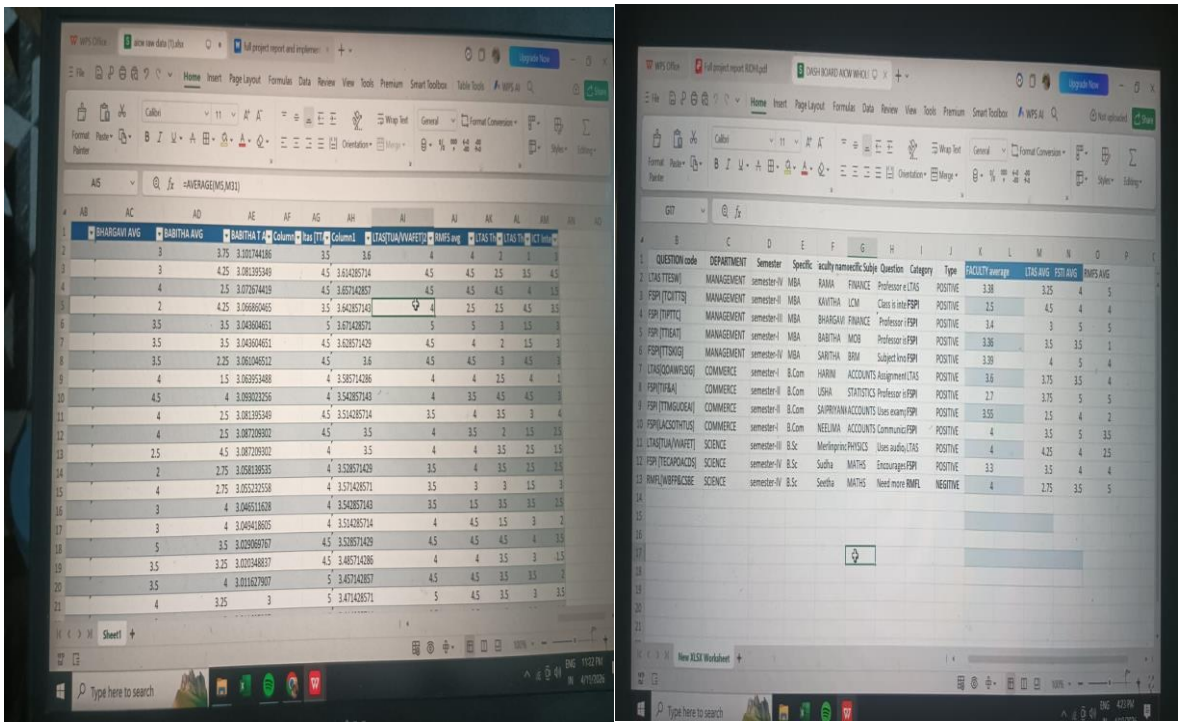
- **Sentiment Extraction:** "Analyze the 'Actionable Suggestion' column for M.Sc. students. Categorize the tone and extract the top 3 recurring themes related to lab access".

- **Innovation Gap Synthesis:** "Summarize the 'Innovation Gap' for the Commerce department. Compare findings between Semester I and Semester III students"
- .
- **QA Protocol Prompt:** "Review the generated summary against the raw data provided. Identify any mentions of facilities or software not explicitly listed in the dataset to ensure zero hallucinations".

- Analyze the 'Actionable Suggestion' and 'Innovation Gap' columns for all M.B.A students. Categorize the feedback into three key themes (e.g., Curriculum, Infrastructure, or Pedagogy) and generate a summary table showing the sentiment tone (Positive/Negative) for each theme."

Appendix D — Screenshot Gallery

Figure 1: Raw vs. Cleaned Dataset Comparison in Excel.



[DFD Level 0 — Context Diagram]

Flow ID	Source (Entity/Store)	Process	Data Flow Content	Target (Entity/Store)
1.0	**PG&UG Students**	**P1: Data Pre-processing**	Ratings (Subject , Clarity) & Text (Quality of assignment work)	**DS1: Processed Feedback**
2.0	**DS1: Processed Feedback**	**P2: AI Sentiment Analysis**	Actionable Suggestion column text	**Microsoft Copilot (AI Engine)**
2.1	**Microsoft Copilot**	**P2: AI Sentiment Analysis**	Sentiment Score & Tone (Categorized)	**DS1: Processed Feedback**
3.0	**Faculty Database**	**P3: RMFS Alignment**	benefit from more practice problems or case studies. prepared for the mid semester exams based on current teaching and practice	**Logic Engine (Excel/Power BI)**
3.1	**DS1: Processed Feedback**	**P3: Alignment Index Calculation**	Subject Ratings	**Logic Engine (Excel/Power BI)**
4.0	**Logic Engine**	**P4: Dashboard Orchestration**	Calculated Alignment_Index & Normalized_Score	**DS2: Power BI Dashboard (.pbix)**
5.0	**DS2: Dashboard**	**P5: Strategic Reporting**	rmfs-Feedback Gap Matrix	**College Principal / Admin**

Figure 2: DFD Level 1 showing the Data Flow from student input to the Alignment Index.

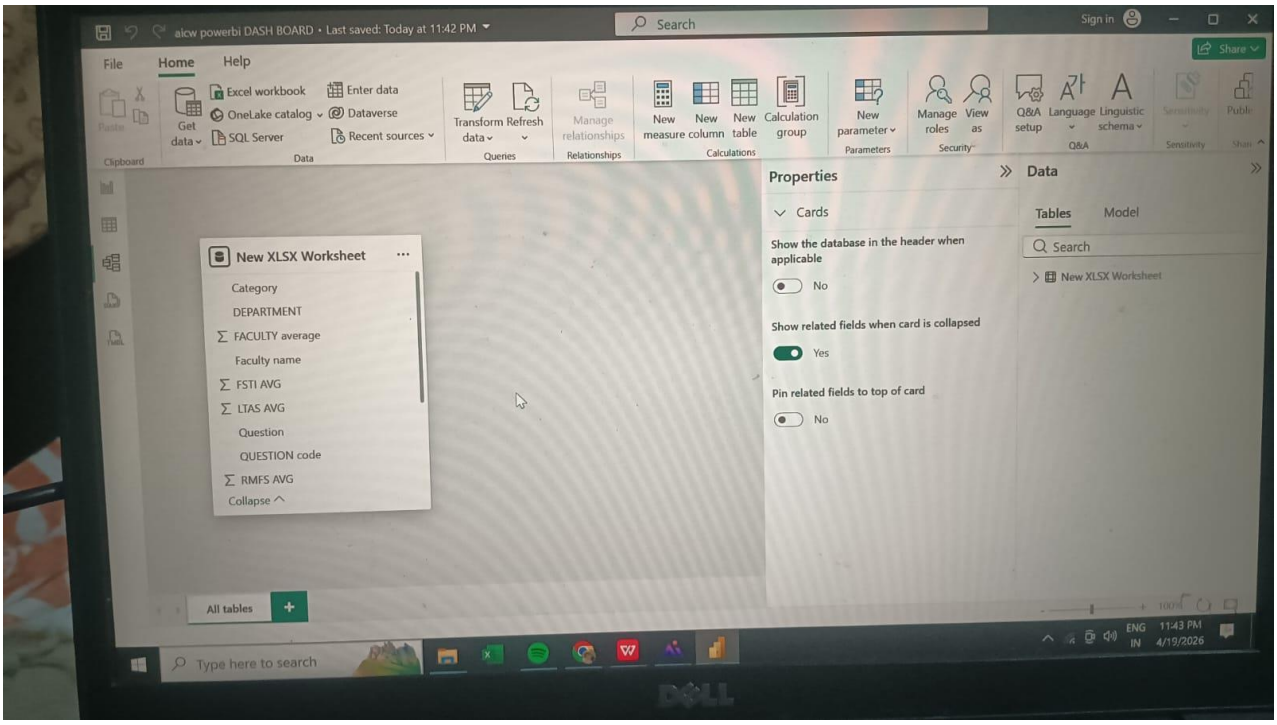


Figure 3: Main Power BI Dashboard featuring the "Feedback Gap Matrix".

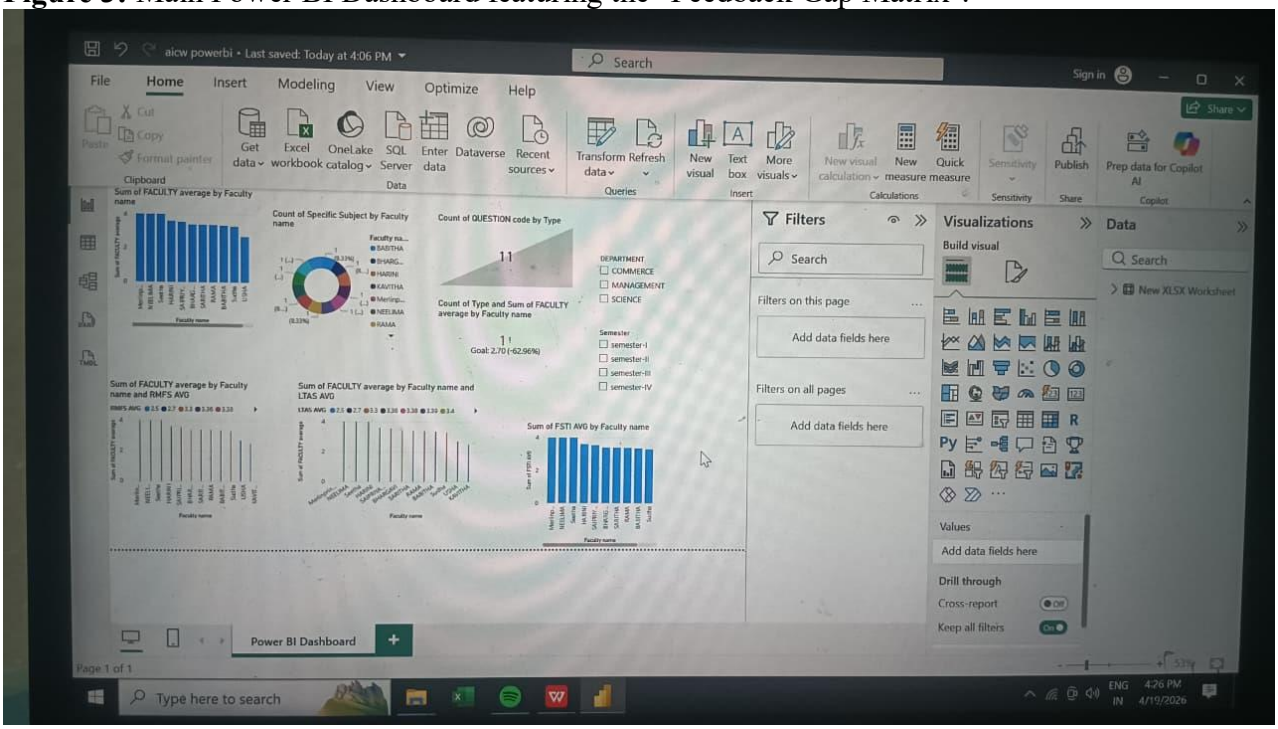


Figure 4: Microsoft Copilot interface during the Sentiment Analysis phase.